Jinhyung Lee, Ph.D.

Postdoctoral Research Associate | Biomedical Engineering | University of Connecticut
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Seeking an Assistant Professor position in Biomedical Engineering

U.S. permanent resident; no visa sponsorship required

Research interests: Biomaterials and nanomedicine; RNA therapeutics; immunoengineering; gene regulation and genome editing.

INDEPENDENT RESEARCH VISION

My independent laboratory will engineer DNA-inspired nanomaterials (called Janus Base Nanomaterial) for RNA delivery and therapeutic immunomodulation. I will study how nanoparticle shape and surface properties influence RNA delivery to immune cells. The program will develop macrophage-targeted RNA therapeutics for rheumatoid arthritis, extend RNA delivery to central nervous system vasculitis, and advance CRISPR-based gene regulation in immune cells.

RESEARCH, FUNDING, AND TRANSLATION

- **Principal Investigator, ORS/Bethel Fellowship ($20,000; 2026–2027).** Evaluating Atsttrin mRNA delivery in patient-derived cartilage-on-a-chip models of osteoarthritis.
  Featured in UConn Today: "Bethel Fellowship Advances UConn Research on Osteoarthritis" (2026).
- **Seven peer-reviewed articles; four as first, co-first, or co-corresponding author.** First author, *PNAS* (2021); co-first author, *Journal of Controlled Release* (2025); co-corresponding author, *Materials Today Advances* (2026).
- **Technology transfer:** Co-inventor of U.S. Patent 12,214,043 B2, licensed by the University of Connecticut to Eascra Biotech.
- **Conference presentations:** Invited podium presentation, BMES 2026 (accepted); podium presentation, ORS 2024.

EDUCATION

Ph.D., Biomedical Engineering — University of Connecticut 2024
Advisor: Dr. Yupeng Chen
Dissertation: *"DNA-Inspired Nanopieces of Drug Delivery in Biomedical Applications"*
M.S., Bioengineering — University of California, San Diego 2017
Advisor: Dr. Yi Chen
Thesis: *"Self-Assembly DNA Nanogel with Embedded Functional Proteins for Nanomedicine Platform"*
B.S., Convergence Biotechnology — Konkuk University 2015

RESEARCH APPOINTMENTS

Postdoctoral Research Associate, University of Connecticut 09/2024–present
PI: Dr. Yupeng Chen. Lead research on targeted RNA delivery for joint disease; collaborate with Dr. Chuan-Ju Liu at Yale School of Medicine on Atsttrin biology.

Collaborate with Dr. Eun Ji Chung at the University of Southern California (USC) on gene editing in the kidney.

Research Assistant, University of Connecticut 08/2019–05/2024
Researcher, Brain Science Institute, KIST 09/2017–12/2018
Graduate Research Assistant, UC San Diego 01/2016–06/2017

TEACHING AND MENTORING

Teaching: Teaching Assistant, Foundations of Biomechanics (BENG 226; graduate level), UC San Diego, 2017; led discussions and developed supplementary materials.

Mentoring: Supervised seven graduate and undergraduate researchers; trainees co-authored four conference presentations.

Teaching interests: Biomaterials; Drug Delivery Systems; Nanomedicine; Tissue Engineering; Introduction to Biomedical Engineering; Gene editing and Gene Therapetuics

PEER-REVIEWED PUBLICATIONS

7 articles | **Lee J** denotes Jinhyung Lee | † Equal contribution | * Corresponding author

1. †Zhai J, †Surian A, Cote T, Zhang W, Dang BT, Wang Y, Chen Q, Hoshino K, ***Lee J**, *Chen Y. "Enhanced endosomal escape for nanoparticle-enabled co-delivery of doxorubicin and siRNA to overcome multidrug resistance." *Materials Today Advances*. 2026;30:100845. doi:10.1016/j.mtadv.2026.100845
2. †**Lee J**, †Zhang W, Nguyen D, Zhou L, Amengual J, Zhai J, Cote T, Landolina M, Ahmadi E, Mishra N, Yu H, Nieh M, *Wang K, *Li Y, *Chen Y. "Computation-aided design of rod-shaped nanoparticles for tumoral targeting." *Journal of Controlled Release*. 2025;387:114169. doi:10.1016/j.jconrel.2025.114169
3. Zhou L, Zhang W, **Lee J**, Kuhn L, *Chen Y. "Controlled self-assembly of DNA-mimicking nanotubes to form a layer-by-layer scaffold for homeostatic tissue constructs." *ACS Applied Materials & Interfaces*. 2021;13(43):51321–51332. doi:10.1021/acsami.1c13345
4. **Lee J**, Sands I, Zhang W, Zhou L, *Chen Y. "DNA-inspired nanomaterials for enhanced endosomal escape." *Proceedings of the National Academy of Sciences USA*. 2021;118(19):e2104511118. doi:10.1073/pnas.2104511118

   Featured in UConn Today: "Escape From the Endosome!" (2021).
5. †Yau A, †**Lee J**, *Chen Y. "Nanomaterials for protein delivery in anticancer applications." *Pharmaceutics*. 2021;13(2):155. doi:10.3390/pharmaceutics13020155
6. Sun M, **Lee J**, Chen Y, Hoshino K. "Studies of nanoparticle delivery with in vitro bio-engineered microtissues." *Bioactive Materials*. 2020;5(4):924–937. doi:10.1016/j.bioactmat.2020.06.016
7. Sands I, **Lee J**, Zhang W, *Chen Y. "RNA delivery via DNA-inspired Janus base nanotubes for extracellular matrix penetration." *MRS Advances*. 2020;5(16):815–823. doi:10.1557/adv.2020.47

PATENTS AND TECHNOLOGY TRANSFER

1. Chen Y, **Lee J**, Zhang W. "Therapeutic nanomaterials." U.S. Patent 12,214,043 B2. Issued February 4, 2025. Licensed by the University of Connecticut to Eascra Biotech.
2. Chen Y, Zhang W, **Lee J**. "Nanomaterial delivery vehicle and method of use thereof." U.S. Patent Application Publication 2025/0302762 A1. Published October 2, 2025; patent pending.
3. **Lee J**, Seong J, Kim H, Choo H. "FRET-based serotonin 2A biosensor." Korean Patent 10-2176538. Granted 2020.

BOOK CHAPTERS

1. Sands I, **Lee J**, *Chen Y. "Nanotechnology for DNA and RNA delivery." *Nanomedicine: Technologies and Applications*, Chapter 8. Elsevier, 2023. doi:10.1016/B978-0-12-818627-5.00008-7
2. **Lee J**, *Chen Y. "Nucleic acid delivery systems for gene therapy." *Gene Delivery: Methods and Applications*, Chapter 2. Nova Science Publishers, 2019.

SELECTED PRESENTATIONS AND PROFESSIONAL SERVICE

1. **Lee J**, Cote T, Zhai J, Dang BT, Chung EJ, Landolina M, Chen Y. "Rod-shaped DNA-inspired Janus base nanoparticles enable in vivo kidney CRISPR editing." **Invited podium presentation**, BMES Annual Meeting, Orlando, FL, October 2026 (accepted).
2. **Lee J**, Cote T, Davidson N, Zhai J, Zhang W, Morgan K, Liu C, Laurencin C, Chen Y. "Janus base nanopieces delivered mRNA therapeutics for osteoarthritis treatment." **Podium presentation**, Orthopaedic Research Society Annual Meeting, 2024.

Additional invited talks: Chonnam National University Medical School (2026); Biomaterials and Biofabrication Conference (2025); Kyung Hee University College of Medicine (2024); Next-Generation Nanomedicine (2023).
Journal reviewer: *ASSAY and Drug Development Technologies* (2026).

Complete CV and references available upon request.